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Evidence of a rheotactic component in the odour search behaviour of freshwater eels

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The detection of food odour by the freshwater eels, *Anguilla australis* and *Anguilla dieffenbachii* released a behavioural response to flow that resulted in direct upstream movement toward the odour source. Changes in various orientation parameters were observed as eels neared the source. Eels had substantially lower swimming velocities and considerably more variable heading and course angles close to the odour source ($\leq$0·9 m) than further away (>0·9 m). Observed changes in orientation parameters were primarily due to changes in the behaviour of a searcher following odour loss. Cross-stream movements were initiated when the eel moved beyond the lateral margins of the odour plume. The behavioural switch from odour-conditioned rheotaxis to cross-stream casting following odour loss occurred more frequently close to the odour source where the plume was most narrow. Odour-conditioned rheotaxis enables the searcher to move quickly and efficiently toward the odour source without the need to extract directional information from a highly intermittent and complex chemical signal.

Key words: cross-stream casting; odour plume; odour search behaviour; odour-conditioned rheotaxis.

INTRODUCTION

In the aquatic environment chemical signals are used to locate prey and food resources. Chemical stimuli have two major advantages over visual, mechanical or acoustic stimuli, they last well beyond the moment of initial production and are extremely specific (Atema, 1980). Accordingly, animals that are nocturnal or reside in visually limited habitats (*e.g.* caves and the deep-ocean) often rely heavily on the sense of olfaction. Although chemical signals possess these advantages, they also have inherent difficulties as the instantaneous distribution of odour or its exact evolution through time is unpredictable (Webster & Weissburg, 2001). Various studies have demonstrated this effect, illustrating that the properties of fluid motion structure the 'odour signal' within the odour plume (Moore & Atema, 1991; Zimmer-Faust *et al.*, 1995; Webster & Weissburg, 2001). These studies have demonstrated that turbulent mixing causes the concentration of odour downstream from the source to be highly variable in time and space.

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With the exception of the studies of Kleerekoper (1978), and Hodgson & Mathewson (1978), the odour search strategies of fishes have received little attention. The most recent reviews of aquatic odour search behaviour (Atema, 1988, 1996; Vickers, 2000; Weissburg, 2000) rely heavily on examples of aquatic benthic crustaceans and flying insects, giving only passing reference to fishes. The orientation mechanisms that aquatic benthic crustaceans use to navigate toward an odour source are reasonably well understood. Laboratory and field studies of the odour search behaviour of the estuarine blue crab *Callinectes sapidus* show that a combination of rheotactic and chemotactic orientation mechanisms are used to localize the source (Weissburg & Zimmer-Faust, 1993, 1994; Zimmer-Faust *et al.*, 1995). Blue crabs employ what is termed an odour-conditioned rheotaxis to move toward the source, relying on flow as an orientating stimulus (Weissburg, 2000). Orientation appears to be the product of two mechanisms: an odour-conditioned response to flow and spatial comparisons of odour distribution. These two mechanisms enable the animal to maintain contact with the plume. Turning or cross-stream movements are only initiated if the searcher exits or loses contact with the plume (Zimmer-Faust *et al.*, 1995).

Although the odour search behaviour of fishes is not well documented, Kleerekoper (1978) and Hodgson & Mathewson (1978) identified two types of odour search strategies: (1) a chemotactic search in which the distribution of the chemical signal appeared to guide the fishes toward the odour source (*i.e.* the chemical signal plays a direct guidance role) and (2) an odour-conditioned rheotaxis in which the detection of odour resulted in the fishes swimming directly upstream in the general direction of the odour source (*i.e.* the chemical signal initiates movements that depend on non-chemical stimuli).

Recently, video and tracking observations of deep-sea and coastal fish species have demonstrated that fishes frequently approach an odour source downstream of the release point, thus highlighting the importance of flow in the odour search strategy of fishes (Dayton & Hessler, 1972; Hodgson & Mathewson, 1978; Emanuel & Dodson, 1979; Wilson & Smith, 1984; Løkkeborg, 1998).

The purpose of this study was to investigate the behavioural mechanisms used by freshwater eels to localize an odour source in a semi-natural field setting and determine whether a rheotactic component is incorporated in odour search behaviour. New Zealand has two species of freshwater eel the shortfin *Anguilla australis* Richardson and the longfin *Anguilla dieffenbachii* Gray. Morphological differences between the two species are minor, with the most obvious distinguishing features being head shape, mouth gape and the length of the dorsal fin (McDowall, 1990). Both species have similar habitat preferences (Jowett & Richardson, 1995) and are nocturnal feeders (Jellyman, 1989). As they grow both species become progressively more piscivorous with shortfin eels >50 cm total length (L_T) and longfin eels >40 cm L_T being almost exclusively piscivorous (Ryan, 1986; Hayes & Rutledge, 1991). Species of the genus *Anguilla* are generally regarded as having to rely heavily on olfactory cues, particularly in feeding (Tesch, 1977).

MATERIALS AND METHODS

Experiments were conducted in an outdoor flow through concrete raceway (5·0 m long × 2·4 m wide) situated at the Silverstream Research Station, Christchurch, New Zealand. The raceway was supplied with natural stream water (16° C) drawn from the surrounding Silverstream system. The raceway emptied directly back into the neighbouring stream. Naturally foraging shortfin and longfin eels residing in the surrounding stream system were free to enter and exit the raceway. This approach was preferable to working with captive animals and enabled quantitative observations of odour search behaviour in free ranging eels.

Three stainless steel mesh screens (1·1 mm^2 mesh size) positioned at 1·3 m (two) and 1·7 m (one) downstream of the entry section served as collimators or flow 'straighteners'. Average flow velocity measured at the cross-sectional centre of the working section was $2 \cdot 1 \pm 0 \cdot 5\ cm\ s^{-1}$. To simulate a natural streambed, the floor of the raceway was covered with cobbles ($5 \cdot 2 \pm 1 \cdot 2$ cm, $n = 20$), obtained from the neighbouring stream. Cross-stream water depth varied from *c.* 41 cm at the centre to *c.* 25 cm at the lateral margins.

STIMULUS PREPARATION AND RELEASE

The odour stimulus for all behavioural observations consisted of 100 g (wet mass) l^{-1} of sockeye salmon *Oncorhynchus nerka* (Walbaum) parr. Fish were decapitated posterior to the pectoral fin insertion and macerated in 1 l of stream water. The solution was then drawn off and filtered to remove any particulate matter. The stimulus was prepared as a single batch and frozen to serve the entire experimental period. Immediately prior to use the stimulus was diluted with natural stream water to achieve a final ratio of 1 : 10 (1 part odour stimulus : 10 parts stream water).

The stimulus was gravity fed via polyethylene tubing (2 mm internal diameter), connected to a blunt syringe needle (1·1 mm internal diameter) bent at right angles. The nozzle of the delivery tube was suspended at the cross-sectional centre of the raceway, *c.* 3 cm above the substratum. The odour stimulus was released at a rate of $52 \pm 2\ ml\ min^{-1}$ (mean ± S.D.; Reynolds number, Re = 1047) and monitored via an inline flowmeter.

STIMULUS MEASUREMENT

To assess the spatially variability of the odour stimulus, odour plumes were visualized by releasing Rhodamine B (Sigma Chemical Co.) under the same conditions as the odour stimulus and video taped. A video capture card and PC were then used to capture a single frame every second. Single frames were converted into three dimensional ($m \times n \times$ [R-G-B]) arrays within MatLab 5·1, with m and n representing the x and y co-ordinates, and the third input representing the true-colour image index. A 30 s (2-D), 'mean-odour-plume' was then constructed by averaging 30 successive image arrays. Under this regime the variability in the distribution of odour was revealed as spatial variations in the intensity of Rhodamine B. The raceway background was then subtracted from mean-odour-plume images, thereby leaving only the Rhodamine stained odour plume. The lateral margins of the plume (defined by the contrast difference between pixels) were then overlaid onto plotted search paths.

ODOUR PLUME STRUCTURE

The resulting odour plume was structured into two distinct regions: a narrow proximal cone and a more distal patch field. The proximal cone region was characterized by well-defined lateral boundaries and bilateral symmetry, these features persisted 0·4–0·7 m downstream of the release point [Fig. 1(a)]. As the plume progressed downstream it lost bilateral symmetry and the lateral margins became noticeably less defined. Mean-odour-plumes derived from time averaged imaging revealed that the intensity of Rhodamine B varied with distance from the source. Within the proximal cone region of the plume intensity remained relatively uniform. Further downstream intensity was considerably more variable at the lateral margins of the plume; this was visible as a graded decrease in intensity from the mid-line toward the lateral margins [Fig. 1(b)].

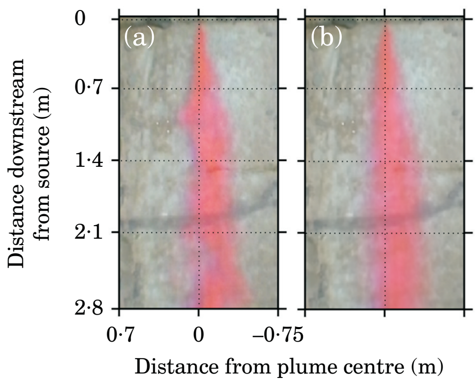

FIG. 1. Odour plumes as visualized with Rhodamine B. (a) Instantaneous distribution and (b) mean odour plume averaged over a 30 s period. Water flow is from top to bottom.

BEHAVIOURAL OBSERVATIONS AND PROCEDURE

Experiments were conducted between 2100 and 0300 hours. Search behaviour was filmed with a SONY CCD camera equipped with a ×0·6 wide-angle lens suspended 3·5 m above the raceway. An *x-y* scale was placed on the floor of the raceway and used to measure absolute distance. The working section was illuminated by two far-red light sources mounted *c*. 2·0 m above the raceway at either end.

SEARCH PARAMETERS AND DATA ANALYSIS

Individual search paths were constructed by capturing single frames every second. Successive frames were imported into NIH Image (Scion Corp.) and an array of *x-y* co-ordinates generated for each individual search path, with the snout of the eel serving as a point of reference. From these digitized paths, several search parameters were analysed; these included swimming velocity, heading angle, course angle and an index of path straightness.

Swimming velocity was measured as the distance travelled (cm) per unit time (s). This measure did not take into account the velocity of the stream. Heading angle was defined as the angle between a vector connecting the source and the current position of the eel and a vector connecting the current position and the next position, with an angle of zero pointing directly at the odour source [Fig. 2(a)]. Course angle was similar to heading angle except that the odour source was replaced with a point directly upstream, with an angle of zero pointing directly upstream [Fig. 2(a)]. Heading and course angles measure the accuracy of orientation relative to the odour source and direction of net stream flow, respectively. All angles were calculated in a clockwise fashion (0–360°).

An index of straightness (Hamilton, 1977; Batschelet, 1981) was calculated for each individual search path to provide a numerical measure of path deviation. This index produces a dimensionless number that lies between 0 and 1. A low value is indicative of a curvaceous search path whereas 1 indicates a completely straight search path. This index has also been termed the net-to-gross-displacement ratio (NGDR) by Weissburg & Zimmer-Faust (1994) and the net-to-gross ratio (NGR) by Moore & Grills (1999).

To examine more closely how search behaviour changed with distance to the odour source, several search parameters were analysed over three equal 0·9 m sections of the raceway. These sections were defined as: close, distances $\leq$ 0·9 m from the odour source; intermediate, distances between 0·9 and 1·8 m from the odour source; far, distances between 1·8 and 2·7 m from the odour source.

Mean heading and course vectors for individual search paths were constructed by applying the trigonometric functions method as described by Batschelet (1981). A mean

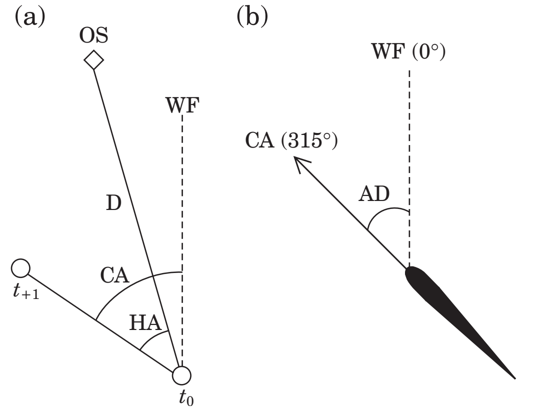

FIG. 2. Description of orientation parameters. (a) Two successive positions of a hypothetical search path are represented by t_0 and t_{+1}. D indicates distance from the odour source (OS) from which heading angle (HA) is calculated. WF indicates the direction of net stream flow from which course angle (CA) is calculated. Swimming velocity is determined by the time between t_0 and t_{+1}. (b) The angular distance (AD) is defined by the angular difference between the assumed *a priori* course of 0° and the eel's actual course angle. In this example the eel has adopted a course angle of 315°, the angular distance then equates to 45°.

angular direction (heading and course) and a mean vector length (r) were calculated for each of the three 0·9 m sections, for each individual search path. The mean vector length provides a measure of variation around the mean angular direction; it lies between 0 and 1 and decreases as the dispersion of data around the mean angular direction increases.

The time spent searching in each of the three 0·9 m sections was calculated for each individual search path. A repeated measures ANOVA was then used to compare whether the time spent searching in each section was similar; this was then followed by a Tukeys *post-hoc* comparison test.

To analyse whether angular orientation parameters varied with distance to the odour source, heading and course angles occurring within each of the three 0·9 m sections were grouped from all individual search paths. Multi-sample testing of angular distances was then used to reveal whether the accuracy of orientation was similar over the three 0·9 m sections. Angular distances were calculated as the shortest angular distance between two points on a circle and reflects the difference between the heading or course angle adopted by the eel and a vector either pointing directly towards the stimulus source or directly upstream, respectively [Fig. 2(b)]. A heading and course angle of 0° towards the odour source and direction of net stream flow was assumed *a priori*. Analysis of angular distances reveals more about search behaviour as it determines departures from a specified angular direction. A two-way Kruskal–Wallis tied ranks test (Zar, 1999) was then employed to compare whether angular distances between each of the three 0·9 m sections were similar; this was followed by a *post-hoc* comparison using Dunn's test (Dunn, 1964).

RESULTS

LOCALIZATION SUCCESS

A total of 31 search paths were obtained over a 3 week period. Due to the slight morphological differences between *A. australis* and *A. dieffenbachii*, it was not possible to identify the species from video observations with any degree of certainty. Eel L_T, as measured from video images, ranged from 45·0 to 60·8 cm (mean ± s.e. = 55·88 ± 1·56 cm, $n = 31$). In each instance the eel successfully located the odour source. Attempts to obtain controls, by substituting the odour stimulus with stream-water, were unsuccessful. The absence of positive responses to 'blank plumes' demonstrated that eels entering the raceway were responding to the odour plume rather than randomly foraging through the raceway.

QUALITATIVE DESCRIPTION OF SEARCH BEHAVIOUR

Orientation behaviour was extremely consistent across the 31 search paths obtained. Thirty of the 31 (97%) search paths featured sustained periods of direct upstream movement toward the odour source, which appeared to be conditional upon the eel remaining within the lateral margins of the odour plume (Fig. 3). In only one search path were large cross-stream movements initiated [Fig. 3(b)]. In this single instance switching back and forth across the current resulted in a meandering search path and turning was loosely associated with crossing or nearing the lateral margins of the plume. As eels closed ($\leq$0·4 m) on the odour source they frequently swam beyond the lateral margins of the odour plume. After exiting the plume eels ceased upstream movement and engaged in 'casting' (cross-stream movements) type behaviour [Fig. 3(c), (d) and (f)]. 'Casts' were most often approximately perpendicular to the direction of net stream flow and severely impeded upstream progress toward the odour source.

ORIENTATION PARAMETERS

Search time

Time taken to localize the odour source ranged from 21 to 51 s (mean ± s.e. = 33·06 ± 0·55 s, $n = 31$). Eels did not partition their total search time equally across the three 0·9 m sections of the raceway (Fig. 4: repeated measures ANOVA: $F_{2,90} = 28{\cdot}62$, $P < 0{\cdot}005$). Eels spent almost twice as much time searching close ($\leq$0·9 m) to the odour source than at either far (Tukeys test, $P < 0{\cdot}005$) or intermediate (Tukeys test, $P < 0{\cdot}005$) distances from the source. Given these results it appears that orientating in the narrow proximal cone region of the odour plume presents the greatest challenge.

Swimming velocity

Mean swimming velocities ranged from 7·77 to 13·84 cm s^{-1} (mean ± s.e. = 10·80 ± 0·32 cm s^{-1}, $n = 31$). Swimming velocity was calculated within 10 cm bins to analyse whether swimming velocity remained constant with distant to the source (*e.g.* 0–10 cm, 10–20 cm) (Fig. 5). The lower swimming velocities seen within 0·4 m of the odour source demonstrated that cross-stream 'casting' movements were substantially slower than direct upstream movements, which occurred at greater distances (>0·9 m) from the odour source.

Search path straightness

Individual search paths typically showed a high degree of straightness and consistency. Straightness indices ranged from a maximum of 0·962, resulting from a practically linear search path [Fig. 3(a)], to a minimum of 0·621, the result of a more curvaceous search path [Fig. 3(b)], in which large cross-stream movements were initiated. The high straightness indices (mean ± s.e. = 0·785 ± 0·020,

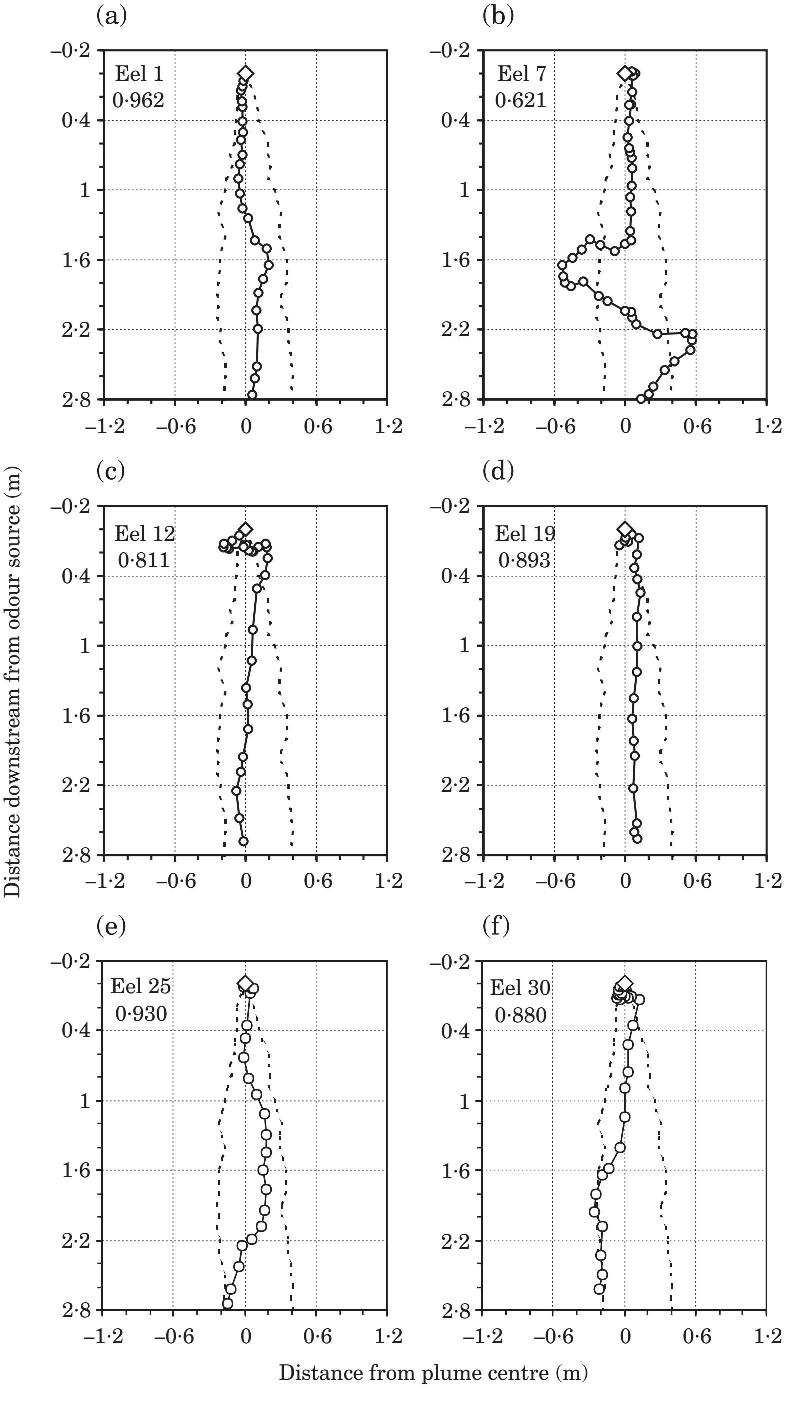

FIG. 3. Six representative search paths (a)–(f) of eels localizing the odour source. ○, positions of individuals at 1 s intervals as they search for the odour source (◇). - - -, the lateral margins of the mean-odour-plume. The straightness index (see text for explanation) is indicated in the top left corner of each search path. Stream flow is from top to bottom.

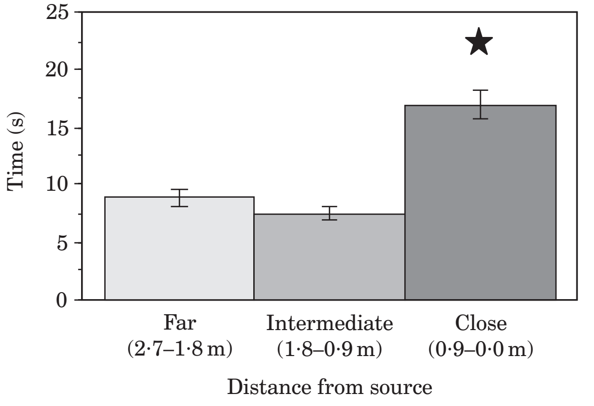

FIG. 4. Time spent searching (mean ± S.E.) with distance to the source. $n = 31$ for each bar. ★, significantly different mean (repeated measures ANOVA, Tukeys test: $P < 0{\cdot}005$).

$n = 31$) observed in this study resulted from sustained periods of direct upstream movement at far and immediate distances (>0·9 m) from the odour source.

Heading and course angle

Changes in mean heading and mean course angles with distance to the odour source for each individual search path are shown in Figs 6 and 7. In all but one case, mean heading and course angles cluster around 0°, regardless of distance to the source. Variability in the vector lengths (r), however, is noticeably different with distance to the source. The greatest variability in mean angular directions (heading and course) as indicated by low vector lengths (r), occurs close (≤0·9 m) to the odour source. Mean heading and course angles are considerably less variable at greater distances from the source.

The mean heading and course angles with distance to the odour source, derived by grouping all individual search paths are given in Tables I and II. Although mean heading and course angles were, as expected, close to zero,

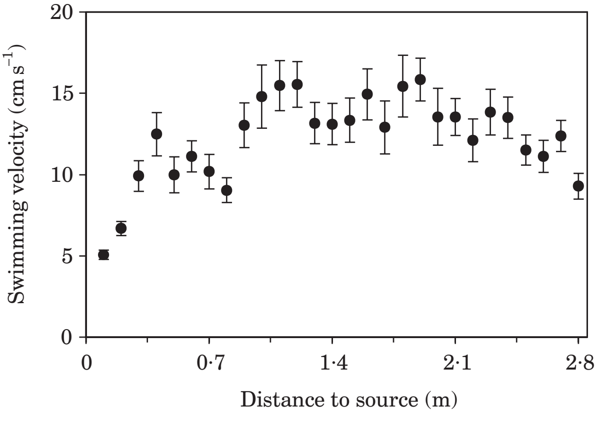

FIG. 5. Swimming velocity (mean ± S.E.), with distance from the source. Means (●) over 10 cm intervals ($n = 1038$).

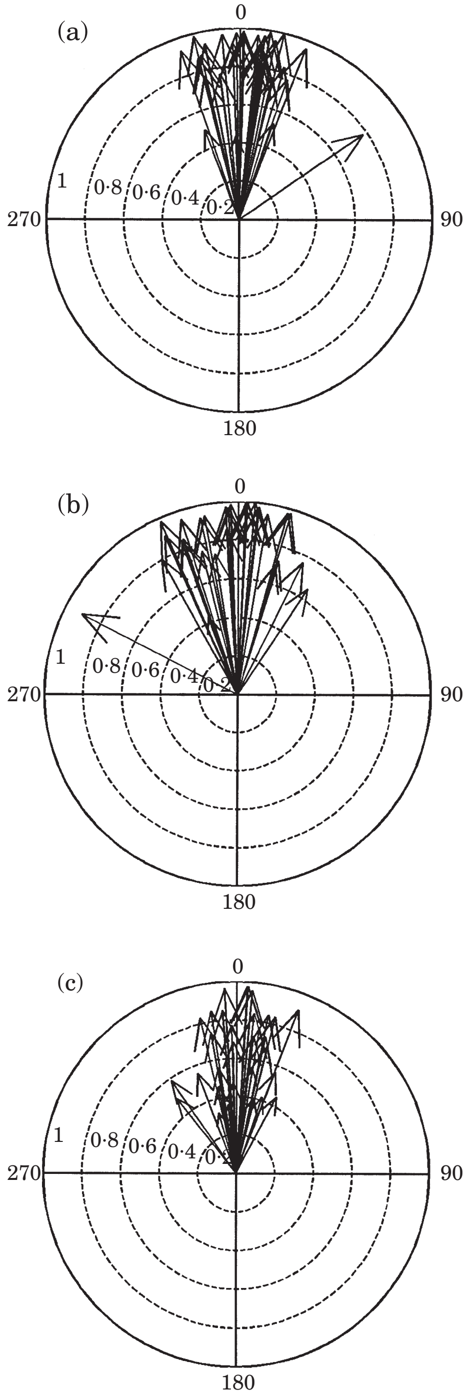

FIG. 6. Circular graphs depicting the mean heading angle and mean vector length (r) for each search path. (a) Far (b) intermediate and (c) close. Grand means for each distance from the odour source: (a) $-359{\cdot}57°$ ($r = 0{\cdot}981$) (b) $-0{\cdot}26°$ ($r = 0{\cdot}987$) and (c) $-2{\cdot}66°$ ($r = 0{\cdot}998$). Grand means were derived using second-order analysis as defined by Batschelet (1981).

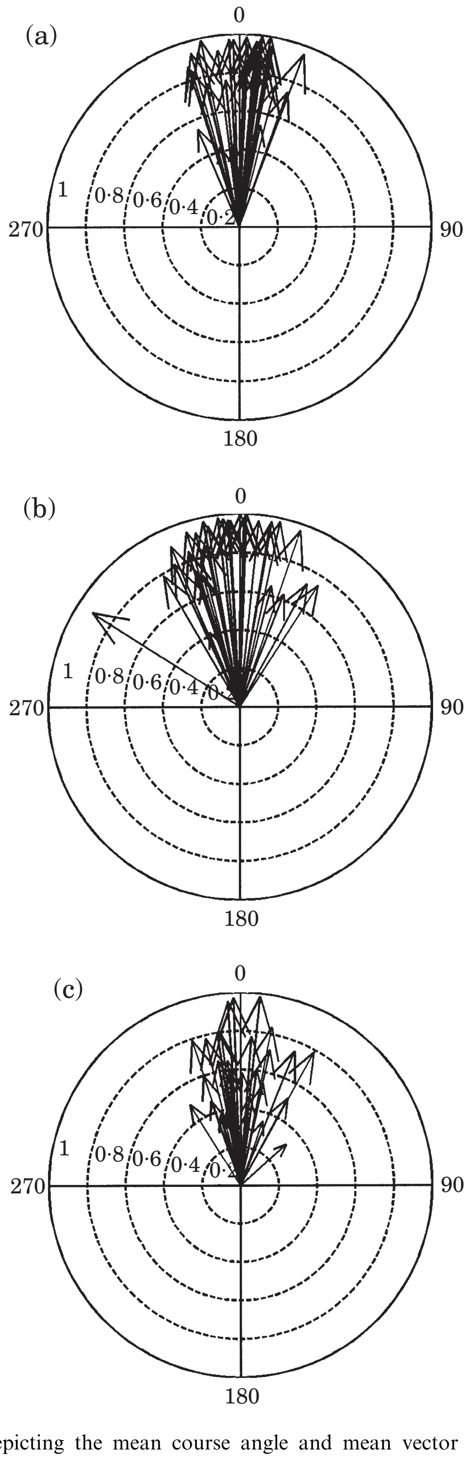

FIG. 7. Circular graphs depicting the mean course angle and mean vector length (r) for each search path. (a) Far (b) intermediate and (c) close. Grand means for each distance from the odour source: (a) $-359{\cdot}47°$ ($r = 0{\cdot}983$) (b) $-359{\cdot}03°$ ($r = 0{\cdot}985$) and (c) $359{\cdot}51°$ ($r = 0{\cdot}861$). Grand means were derived using second-order analysis as defined by Batschelet (1981).

differences in the mean vector length and mean angular deviation further highlighted changes in search behaviour with distance to the source. The mean heading vector length was 23·6 and 25·6% lower within 0·9 m of the odour source with mean angular deviation increasing by 13·16 and 14·87°. A similar trend was also observed with mean course angle, with a 31·8 and 32·2% decrease in mean vector length within 0·9 m of the odour source and a coincident 17·34 and 17·79° increase in mean angular deviation highlighting the variability in the accuracy of orientation with distance to the odour source.

Analysis of angular distances confirmed that the accuracy of orientation, with respect to the odour source and direction of net stream flow varied with distance from the odour source. Eels did not orientate with equal accuracy to either the odour source (two-way: Kruskal–Wallis test, $P < 0.005$) or direction of net stream flow (two-way: Kruskal–Wallis test, $P < 0.005$) over the three 0·9 m sections. Departures from the specified angular heading and course direction of 0° were significantly greater close ($\leq$0·9 m) to the odour source than at either far (Dunn test, $P < 0.005$: for both heading and course) or intermediate (Dunn test, $P < 0.005$: for both heading and course) distances. Differences in angular distances close $\leq$0·9 m to the odour source evidently arise from cross-stream 'casting' behaviour, which primarily occurred within 0·4 m of the source. Departures from the specified heading and course angle of 0° were not significantly different between far and intermediate distances (Dunn test: NS: for both heading and course).

DISCUSSION

The observations suggest that odour search behaviour in freshwater eels is principally based upon two orientation mechanisms: an odour-conditioned rheotaxis and cross-stream 'casting'.

TABLE I. Mean heading angle, vector length (r) and angular deviation across all 31 search paths. n, the number of digitized points

Distance from source (m)	Mean vector (°)	Mean vector length (r)	Mean angular deviation (°)	n
2·7–1·8	1·58	0·770	38·86	272
1·8–0·9	355·47	0·790	37·15	243
0·9–0·0	0·76	0·588	52·02	523

TABLE II. Mean course angle, vector length (r) and angular deviation across all 31 search paths. n, the number of digitized points

Distance from source (m)	Mean vector (°)	Mean vector length (r)	Mean angular deviation (°)	n
2·7–1·8	7·33	0·779	38·09	272
1·8–0·9	355·69	0·784	37·64	243
0·9–0·0	0·81	0·532	55·43	523

Initial detection of a food odour releases a conditioned response to water flow resulting in direct upstream movement toward the odour source. The rheotactic component of search behaviour is most evident in the high straightness indices and low variability in mean heading and course angles at distances >0·9 m from the odour source (Figs 6 and 7). Cross-stream casting movements were only initiated when the searcher lost contact with the odour plume, which typically occurred within 40 cm of the odour source. The switch to 'casting' behaviour following odour loss is evident in the changes in several orientation parameters between 0·9 and 0·0 m from the odour source: low mean heading and course vector lengths (Figs 6 and 7), greater amount of time spent searching (Fig. 4) and considerably lower swimming velocities (Fig. 5). The behavioural switch from odour-conditioned rheotaxis to cross-stream 'casting' clearly contributed to final localization of the source.

Field studies using fixed baited stations have frequently shown that fishes consistently approach an odour source downstream of the release point (Dayton & Hessler, 1972; Wilson & Smith, 1984; Løkkeborg *et al.*, 1989; Løkkeborg, 1998) and highlights the fact that flow direction can be used as a reliable orientating stimulus when searching for the source of an odour. By acoustically monitoring the movements of cod *Gadus morhua* L. toward a baited longline, Løkkeborg (1998) was able to show that following the detection of odour cod swam several hundred metres directly up current toward the odour source. Using video observations, Montgomery *et al.* (1999) documented similar search behaviour in *Trematomus bernacchii* Boulenger, with fish consistently approaching the odour source downstream of the release point with relatively straight search paths. By comparing actual search paths with computer simulations, Montgomery *et al.* (1999) suggested that search paths resulted from the integration of both rheosensory and chemosensory information. The odour search behaviour documented in this study closely parallels these previous observations, demonstrating that for these fishes at least, the direction of the flow dispersing the odour can provide crucial information about the location of the odour source.

The reliance on flow as an initial orientating stimulus is not a behavioural strategy unique to fishes but is a consistent element in the orientation strategies of many aquatic invertebrates (Bell & Tobin, 1982; Brown & Rittschof, 1984; Weissburg & Zimmer-Faust, 1993, 1994; Zimmer-Faust *et al.*, 1995; Zhou & Rebach, 1999). Because flow disperses the odour from the source, the flow itself can provide the searcher with an additional piece of information, providing what could be termed a directional 'signpost' to the source. Resolving flow direction allows the searcher to determine what course to take towards an odour source that is often well beyond visual range (Løkkeborg, 1998; Montgomery *et al.*, 1999; Vickers, 2000).

Freshwater eels appear dependent on two cues to locate the source of the odour plume. The direction of stream flow is essentially used as a general orientation cue, in essence providing the searcher with an approximate direction in which to move. Eels consistently turned back towards the plume after they exited it, in this case it seems most likely that eels are also employing some temporal or spatial odour-sampling scheme to localize the source. This observation is supported by the fact that 'casting' was never randomly orientated, following odour loss eels consistently 'cast' in the direction that took them back towards the plume. Therefore, eels appear to have the ability to detect and track the lateral margins or edges of the odour plume.

Extracting information about the location of the odour source using the chemical signal alone, presents the searcher with two alternatives, either derive several time-averaged samples of odour concentration at multiple locations within the plume or undertake a detailed analysis of certain instantaneous features (*e.g.* onset slope and burst shape) of the odour signal. Webster & Weissburg (2001) have recently demonstrated that a searcher would be required to wait at each discrete location for several minutes to arrive at a reliable time averaged estimate of odour concentration. Given the long sampling periods necessary, this strategy seems totally impracticable for many aquatic animals. Several studies have demonstrated that directional information is contained within various instantaneous features of the odour signal (Moore & Atema, 1991; Zimmer-Faust *et al.*, 1995; Webster & Weissburg, 2001). To extract such information, however, these studies have typically used extremely high temporal sampling rates (10–250 Hz), which maybe well beyond the chemosensory sampling capabilities of aquatic animals (Gomez & Atema, 1996).

It has recently been proposed that odour-conditioned rheotaxis in the estuarine blue crab enables the searcher to quickly and efficiently narrow their search window (Weissburg, 2000), as cross-stream 'casting' movements are only initiated when the searcher loses contact with the odour plume. These observations seem consistent with the present study. The rheotactic component of odour search behaviour consistently enabled eels to get within 40 cm of the odour source before losing contact with the odour plume, resulting in a considerable narrowing of their search 'window'.

A critical decision for animals tracking turbulent odour plumes is when to alter search behaviour following odour loss (Vickers, 2000). Responding to minute gaps in the odour signal would substantially lower search efficiency and not contribute greatly to upstream progress toward the odour source. Conversely, ignoring large gaps would considerably reduce the chances of localizing the odour source for a searcher responding rheotactically and moving upstream may exit the odour plume and continue moving well beyond the source. In this respect it is interesting to look at the latency between switches in a search behaviour following odour loss, which appears most frequently in the region of 1–2 s [Fig. 3(c)–(f)]. This seems to indicate that for freshwater eels a gap of >1 s is required before the searcher will initiate a change in behaviour; similar latencies are seen in estuarine blue crabs responding to odour plumes in a tidal estuary (Zimmer-Faust *et al.*, 1995).

In order to deduce flow direction animals must first reference their movement relative to that of stationary objects in the environment, this is either achieved visually or tactilely. In only one species has it been shown that upstream movement in the presence of an odour stimulus (ovarian fluid) relies upon an optokinetic response (Emanuel & Dodson, 1979). For nocturnally active species or those that live in turbid visually limiting environments, visual determination of flow direction may not be practical. In this instance other sensory mechanisms may be employed to deduce flow direction. For fishes, the mechanosensory lateral line system has been shown to be behaviourally important in the orientation to water currents (Montgomery *et al.*, 1997). Recently, Voigt *et al.* (2000), demonstrated that some receptors of the lateral line system of the New Zealand longfinned eel have non-adapting properties, indicating that these receptors, presumed to be superficial neuromasts, detect absolute flow-velocity. Theoretically, flow direction could be determined by comparing the responses of differently orientated neuromasts over the head and may provide the basis for non-visual determination of water flow direction. Consequently, the lateral line system may make an important sensory contribution to the odour search behaviour of freshwater eels and other fishes in general by providing the fishes with the necessary information needed to resolve flow direction.

Although this study has characterized the odour search behaviour of freshwater eels in a semi-natural turbulent flow environment the limitations of the study must be stressed. This study utilized a jet-type stimulus primarily because plume structure was highly reproducible and it mimicked the stimulus release properties used in other studies of odour search behaviour (Moore *et al.*, 1991; Weissburg & Zimmer-Faust, 1994; Consi *et al.*, 1995; Zimmer-Faust *et al.*, 1995). Jet-type stimuli, however, typically have high exit velocities relative to the carrier flow. This induces a large degree of shear between the stimulus and the carrier flow and results in considerable mixing of the ambient fluid and odour stimulus (Webster & Weissburg, 2001). As Weissburg (2000) mentioned, this type of stimulus release does not mimic the situation of passively leaking objects. Therefore, before being able to draw any firm conclusions about the odour search strategies of fishes, further exploration of search behaviour under a variety of stimulus release conditions is clearly required.

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